

IoT Based Smart Street Light

CNT5006

Mr.Sachintha



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Introduction

The main focus of this presentation is the creation and use of a smart street lighting system based on the Internet of Things, examining its requirements, design and application in modern cities. We will also cover the technical aspects of the intelligent information processing and automatic control systems that make this modern solution possible.

Team Members

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The Imperative for Smart Street Lighting



Energy Efficiency

- Traditional street lights consume a lot of energy; however, smart systems reduce consumption by providing adaptive lighting.

Cost Reduction

- Save significantly on operating costs by reducing energy consumption and maintenance costs with predictive analytics.

Enhanced Urban Living

- Cities are better when the air is cleaner, the streets are quieter, safety is improved, and parks are expanded.

Future-Proofing

- Smart street lights form the essential infrastructure for other smart city applications, thereby expanding the horizons of urban development.

Understanding End-User Requirements



- Accurately gathering system requirements from various stakeholders is essential to create an efficient IoT-based smart street lighting system. The strategy we use is one of several methods for gathering information.

Information Collection Techniques

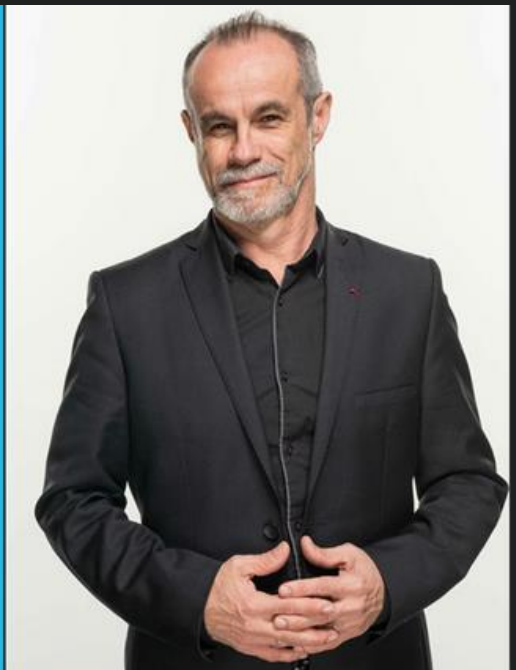
01. Interviews: Engaging with city planners, maintenance crews, and public safety officials.

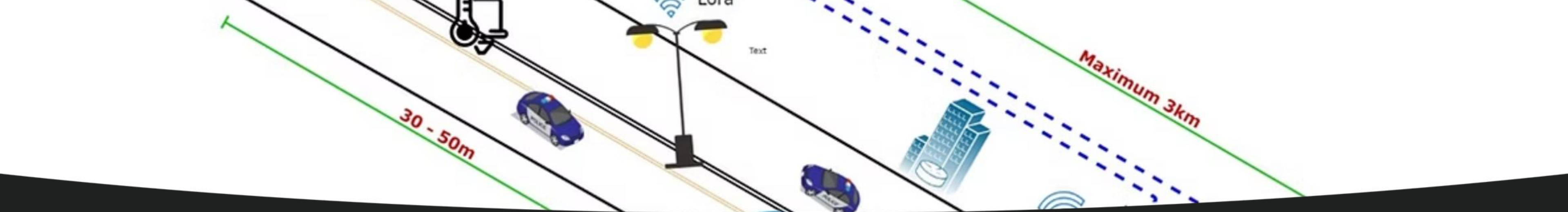
02. Surveys: Gathering quantitative data on current lighting issues and desired features from residents.

03. Observation: Analyzing existing street lighting patterns and energy consumption.

04. Workshops: Collaborative sessions to define functionalities and priorities.

**C40's interview with
Prof Carlos Moreno,
sustainable design and
urban planning expert**





Key Functional Requirements

Based on end-user feedback and system analysis, the following functional requirements are critical for our Smart Street Lighting system.

Adaptive Dimming

Automatic adjustment of light intensity based on real-time conditions.

Motion Detection

Sensors to detect human or vehicular presence, enabling on-demand lighting.

Fault Detection & Reporting

Automated alerts for malfunctions, reducing response times and improving maintenance.

Centralized Control

A central platform for monitoring, managing, and scheduling all street lights.

Non-Functional Requirements: Ensuring Robustness

In addition to critical functions, non-functional requirements specify quality characteristics that are critical to the performance, reliability, and security of the system.

Performance

Real-time response for lighting adjustments and data processing (latency < 500ms).

Security

Robust data encryption and access control to protect sensitive urban data.

Reliability

High uptime (99.9%) and fault tolerance for continuous operation.

Scalability

Ability to integrate thousands of lights and new sensors without degrading performance.

Analysis and Design Plan

The analysis phase translates requirements into a detailed design blueprint, ensuring all technical and operational aspects are considered.

Requirements

Analysis

Detailed breakdown of functional and non-functional needs, creating use cases and user stories.

System Architecture

Definition

Designing the IoT sensor network, communication protocols, cloud platform, and control interfaces.

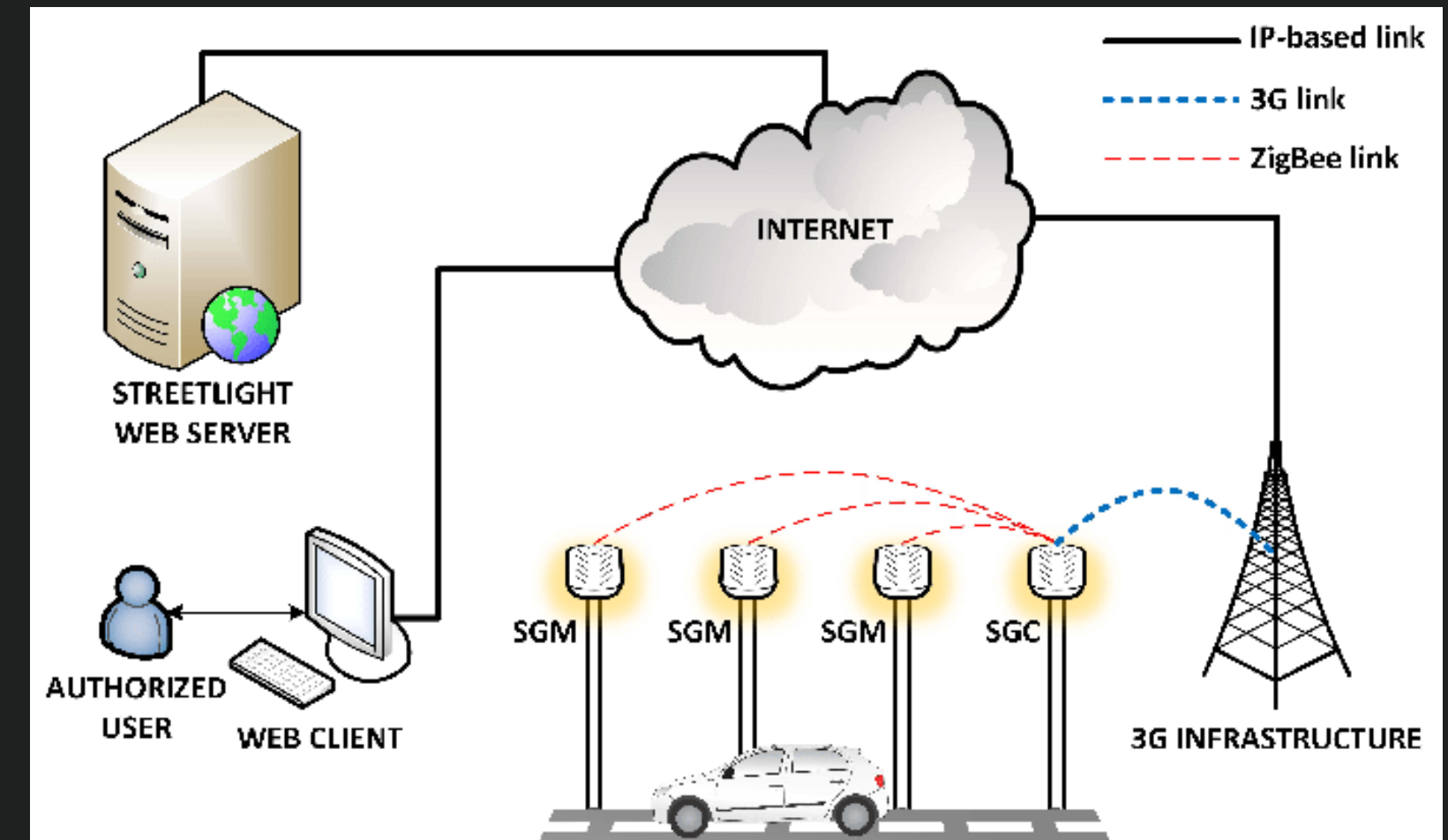
Component Design

Specifying hardware (sensors, controllers) and software modules (firmware, cloud services, dashboards).

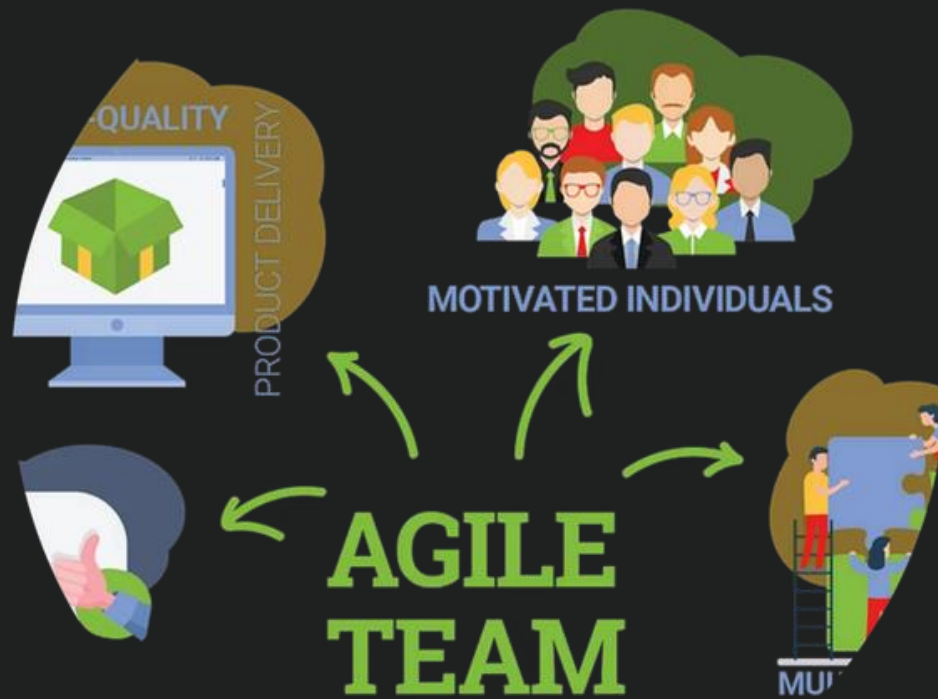
Security & Privacy

Design

Integrating security measures from the ground up, including data anonymization and access control.



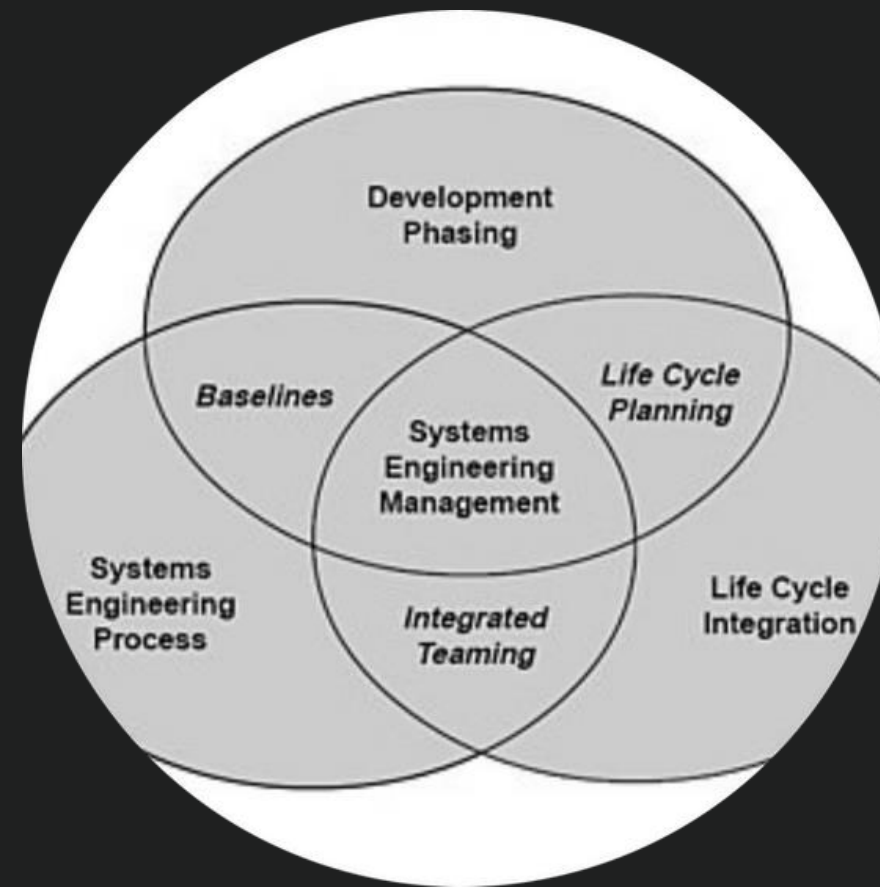
System Design Methodologies



Agile

Development

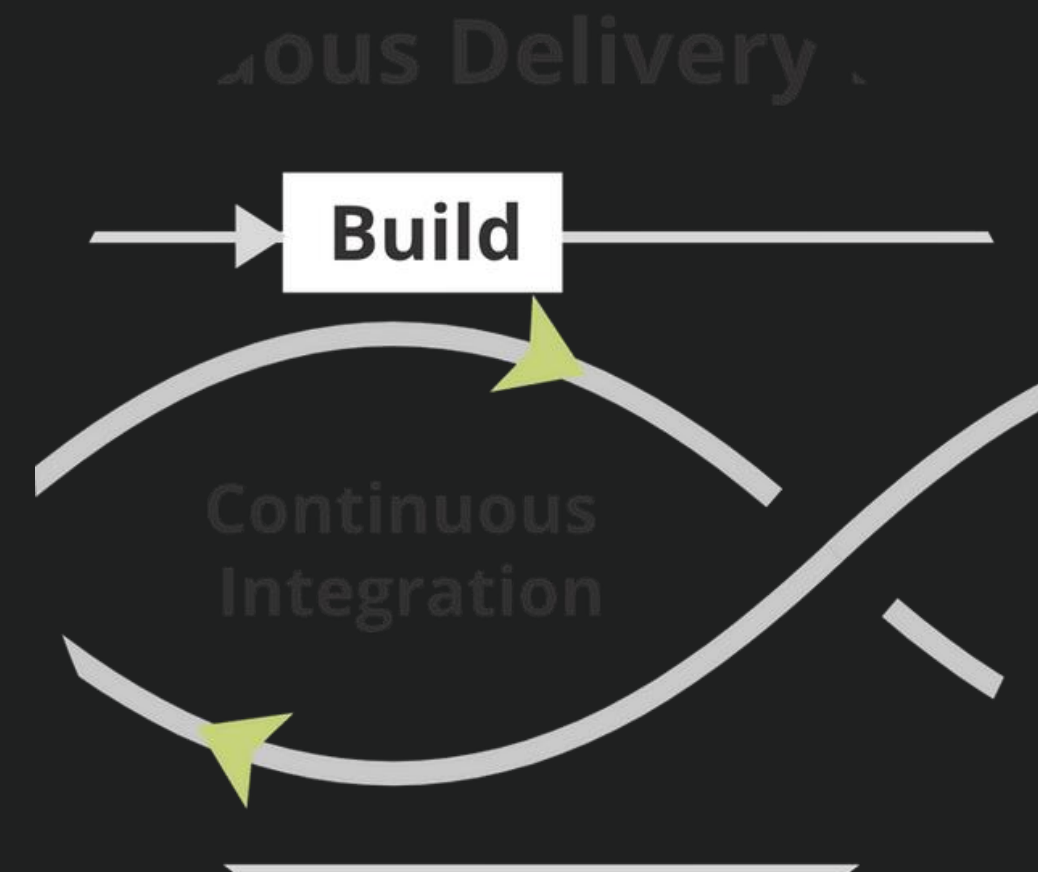
Iterative and incremental approach, allowing for flexibility and rapid adaptation to changing requirements.



Systems

Engineering

Holistic approach focusing on the entire lifecycle, from concept to deployment and maintenance.



DevOps

Principles

Integrating development and operations to shorten the system development lifecycle and ensure continuous delivery.

Essential Tools for Implementation



The selection of robust tools is paramount for developing, deploying, and managing an IoT-based Smart Street Lighting system.

Hardware Prototyping

Arduino/Raspberry Pi for sensor integration and local control unit development.

Cloud Platform

AWS IoT, Azure IoT, or Google Cloud IoT for data ingestion, processing, and analytics.

Communication Protocols

MQTT for lightweight messaging and LoRaWAN for long-range, low-power connectivity.

Programming Languages

Python for backend logic, data processing, and machine learning algorithms.

Intelligent Information Processing & Control Systems

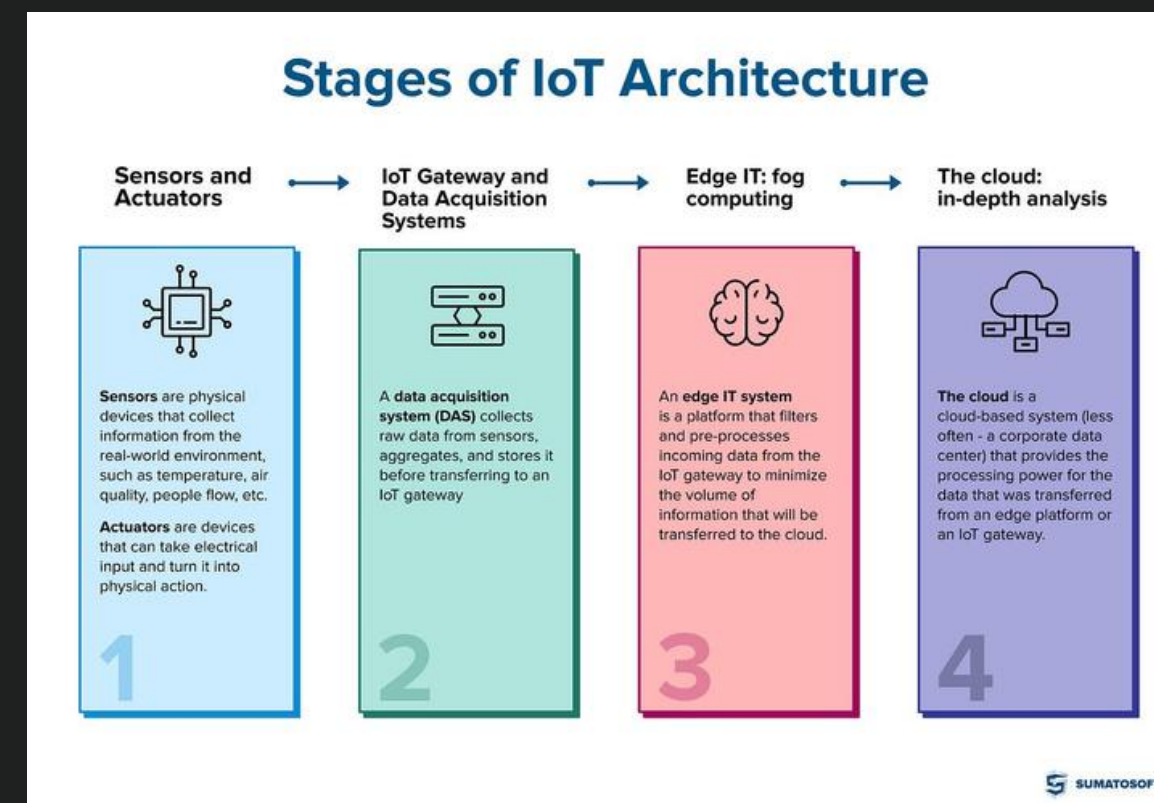
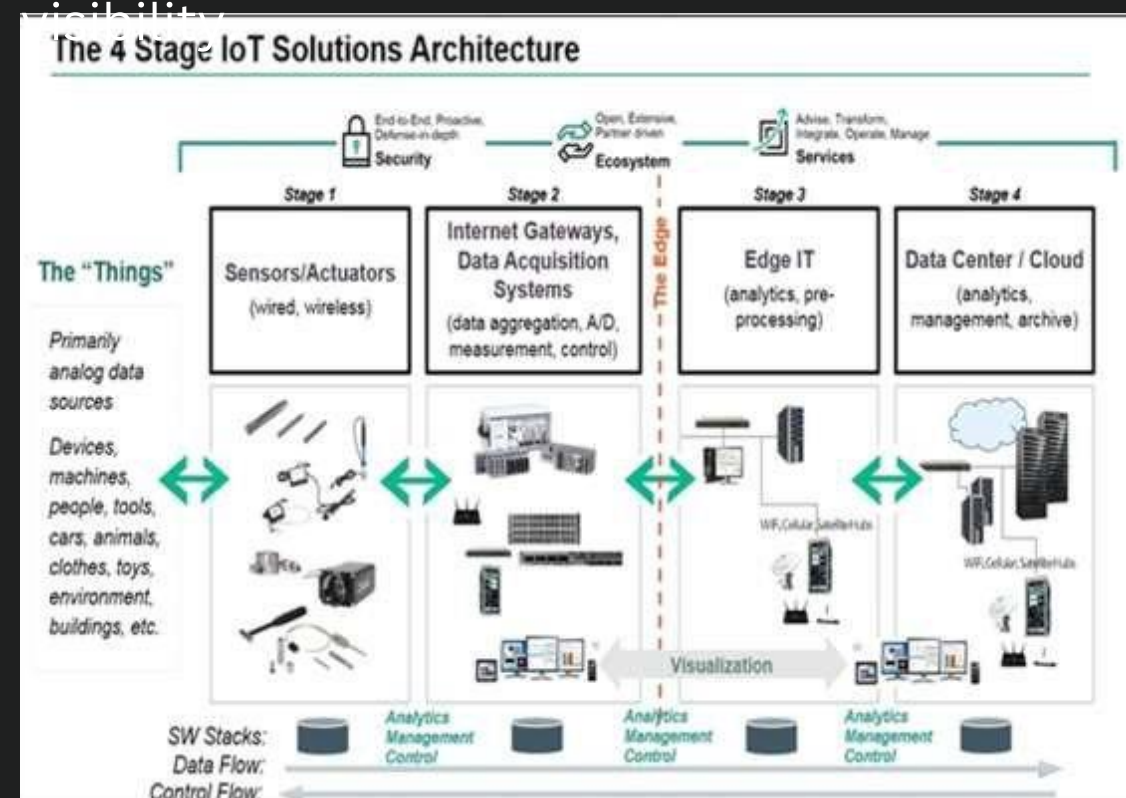
At the core of the Smart Street Lighting system are advanced intelligent processing and automatic control mechanisms.

Data Acquisition: Sensors (light, motion, environmental) collect real-time data.

Edge Computing: Local processing to reduce latency for immediate actions like motion-triggered dimming.

Cloud Analytics: Big data analysis for long-term trends, predictive maintenance, and optimization algorithms.

Feedback Control Loops: Algorithms adjust lighting based on sensor input and predefined policies, optimizing energy and





Conclusion: The Future of Urban Lighting

The IoT-based Smart Street Lighting system is a revolutionary alternative for urban areas. The application of smart information processing and self-governing control mechanisms enables municipalities to experience unmatched energy savings, cost efficiencies, and improved security for the public. The use of this technology goes beyond just illuminating roads; rather, it is a preparation for the future that encompasses the making of smart, eco-friendly, and responsive cities.

**Thank
You**

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